

Cross-Lingual Prototypicality Bias in Multimodal Evaluation Metrics and VLM Judges

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Abstract

Would a vision–language judge give the same answer if the images and intended meaning stayed fixed but the language changed? We test this on PROTOBIAS, where one image is correct but atypical and the other looks typical while violating one detail. Two VLM families judge the same pairs in seven languages, producing 12,596 forced choices. Language matters: 68% of Qwen items and 69% of InternVL items change answer in at least one language, and Bengali and Greek draw the highest error rates while English draws the lowest. Yet we detect no attribute–language interaction: both models stay most vulnerable on wealth by a wide margin and near chance on their least biased attribute. Three independent scorers share the wealth contrast, but multilingual SigLIP ranks the languages differently, placing Bengali lowest where the judges place it highest. Because every language is asked about the identical image pair, these differences cannot be attributed to the images. That makes prototypicality bias a cross-lingual evaluation problem, not only an image-scoring problem.

1 Introduction

Suppose a vision–language model sees the same two images twice. The question means the same thing both times, but once it is asked in English and once in Bengali. If the model is judging image–text fit, should its answer change? This is a natural-language issue inside a visual evaluation task: the evidence is fixed, but the route through language is not.

Roy et al. (2026) show that automatic judges often reward what looks typical rather than what a sentence says. PROTOBIAS pairs a correct but atypical image with a prototypical image that violates one detail. In Figure 1, the futon sits on the grey carpet the sentence asks for; the bed does

not. Both of the judges we test — Qwen2.5-VL-7B (Bai et al., 2025) and InternVL3-8B (Zhu et al., 2025) — answer this correctly in some languages and not in others.

PROTOBIAS establishes the failure in English across several evaluator families. We ask what happens when the linguistic input changes while the image pair stays byte-identical, and whether any language effect changes only the number of errors or also which meanings are vulnerable.

We ask both judges to choose across seven languages, then compare their choices with three image–text scorers. An integrity audit of our own pipeline (§5.1) caught a cache key that could have paired items with the wrong sentence; the results below are from the corrected re-run.

- RQ1** Does changing prompt language change the judges’ answers?
- RQ2** Does language change *which* categories are vulnerable, or mainly the overall error rate?
- RQ3** Do two VLM families fail on the same items?
- RQ4** Do separate image–text scorers show the same category and language patterns?

Our contribution is a controlled cross-lingual audit with fixed visual evidence, item-level rather than only aggregate comparison, and a prompt-integrity audit. Code and data: https://github.com/xchuan-li/cross_lingual_protobias. This is a single-author submission; Appendix A states who did what and Appendix F documents how AI assistance was used.

2 Related work

Automatic judges of image–text alignment. CLIPScore (Hessel et al., 2021) uses scaled, clipped CLIP similarity; PickScore (Kirstain et al.,

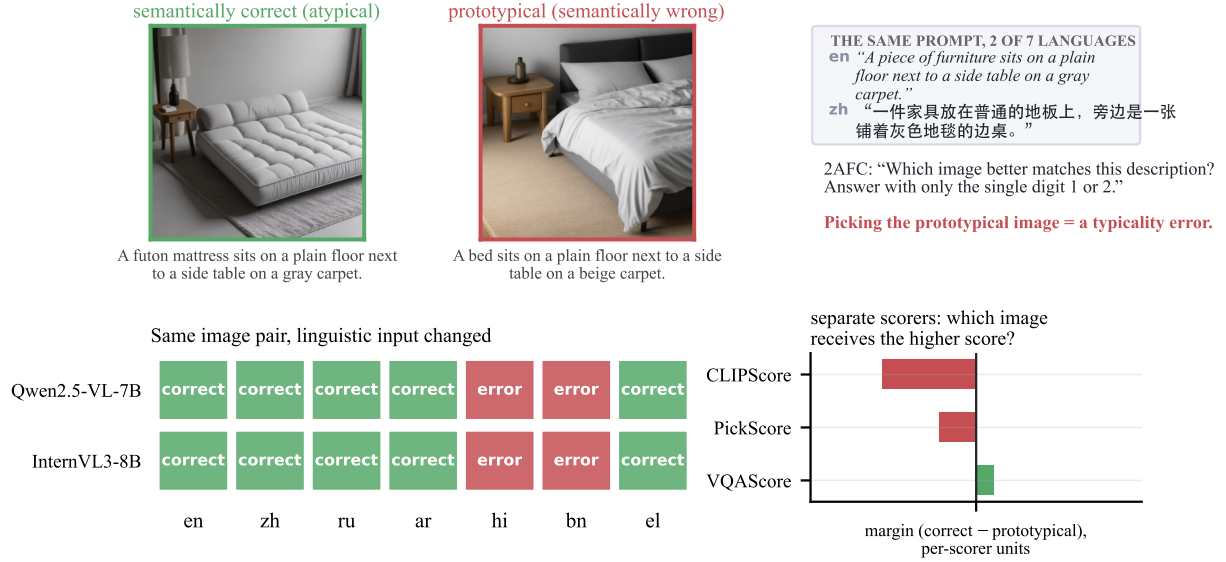


Figure 1: Our running example (PROTOBIAS item 734) and the experimental setting. The neutral prompt uses the hypernym *a piece of furniture* plus one discriminating detail: the carpet is *gray*. Both the description and the instruction around it are translated; the images never change (top right shows two of the seven languages, all instruction templates are in Appendix E). The correct image shows an atypical piece of furniture, a futon mattress, on a grey carpet; the prototypical image shows a canonical bed but puts it on a beige carpet. Bottom left: both judges see this identical pair and only the linguistic input changes, yet the verdict moves with it — both are correct in English, Chinese, Russian, Arabic and Greek and both fail in Hindi and Bengali. Bottom right: without asking a generative VLM to answer, CLIPScore and PickScore score the prototypical image higher. We return to this example in the results.

2023) trains a CLIP-based scorer on user preferences. VQAScore (Lin et al., 2024) uses the probability that a generative VQA model answers “yes” when asked whether an image depicts the text. We use all three as complementary probes.

Prototypicality bias. Roy et al. (2026) define prototypicality bias and introduce PROTOBIAS: correct but less-prototypical images paired with prototypical adversaries that contain one semantic violation, across animals, objects and demography. They show failures in several evaluator families and introduce the lightweight 7B ProtoScore as a mitigation baseline; our results suggest that such a baseline also needs testing across languages and categories. We keep their data but study category structure, cross-lingual modulation and item-level agreement, extending their prototype-theoretic framing (Rosch, 1975).

Judges are biased, and bias measurements are fragile. Human and LLM judges can both be systematically biased (Chen et al., 2024). Bias estimates also vary with instructions, few-shot examples, prompts, metrics and sampling (Hida et al., 2025; Akyürek et al., 2022), and pairwise judges are sensitive to the order in which the candidates

appear (Wang et al., 2024). We therefore fix the instruction, randomise the image slot per trial, and test answer slot and visual cue as confounds (§5.6). Our setting extends these prompt-sensitivity findings to one where the visual evidence is held fixed.

Bias evaluation beyond English. Gupta et al. (2025) study typicality gradients across five languages using human norms and a text-only multilingual LLM, and report cross-language variation. Jin et al. (2025) compare social-bias evaluation in English and Korean. In the multimodal case, Romero et al. (2024) argue that translating questions while reusing the same images gives a narrow view of cultural competence, and answer it by collecting culturally grounded images per region. That critique is well taken for measuring cultural knowledge, but the same property is what our question requires: because the pair is byte-identical across languages, any difference in the verdict has to come from the language rather than from what is depicted. We therefore hold the images fixed, change only the language, and compare multimodal PROTOBIAS decisions with scorer outputs item by item.

domain	subcategory	<i>n</i>	partition	<i>n</i>
animal	animal	91	morality	93
	bird	112	intellect	48
	mammal	97	civility	64
object	furniture	91	power	64
	tableware	100	wealth	31
	vehicle	109	colour	210
demography	nationality	98	count	229
	religion	98	layout	182
	sexual orient.	104	scale	100
			spatial	179
total items		900		

Table 1: The 900 sampled items, by domain and subcategory (left) and by the social attribute or manipulated visual cue (right). Demography items carry an attribute; every item carries exactly one cue.

3 Data

We use PROTOBIAS (Roy et al., 2026), released as `subha-roy/dl4dh_data`: 1,500 items split evenly over animal, object and demography. Each has a neutral prompt and two 1024×1024 images. The prompt names a hypernym (*a bird, a vehicle*) plus a discriminating detail. The correct image satisfies it with an atypical instance; the prototypical image violates it with a canonical instance (Figure 1).

Structure we exploit. Metadata exposes three finer subcategories per domain, five social attributes for demography, and the manipulated visual cue (count, colour, spatial arrangement, layout, or scale). We sample 300 items per domain (seed 42), 900 in all; Table 1 gives the breakdown. The cells are uneven by construction: wealth has only 31 items and scale only 100, so attribute and cue results carry Wilson intervals and the regression pools languages.

Translation. We translate only the neutral prompt into Chinese, Russian, Arabic, Hindi, Bengali and Greek with Google Translate via `deep-translator`; the question templates are language-specific and answers are ASCII 1/2. Images remain byte-identical. A 40-prompt-per-language back-translation audit gives token-F1 of 0.79–0.88 (Appendix C).

Ethics. Demography items encode stereotypes involving religion, nationality and sexual orientation. We show only neutral-domain images and report demography in aggregate. “Prototypical” refers to the benchmark’s contrast, not a fact about people.

4 Method

Each item follows two paths: a VLM chooses between its images once per language, while three automated scorers assign a score to each image–text pair without making the two-image forced choice. Row-index keys join the paths one-to-one.

4.1 Behavioural probe: 2AFC

For each item and language we present both images and the translated prompt and ask for a two-alternative forced choice (Figure 2). The English template is

```
Look at Image 1 and Image 2.
Which image better matches this
description: "{prompt}"?
Answer with only the single
digit 1 or 2.
```

The six other languages use language-specific templates; all seven are given verbatim in Appendix E, together with a back-translation check that they carry the same instruction. We randomise the prototypical image’s slot per (item, language, model) and record it (§5.6). We parse the first 1/2; failures are dropped and counted.

The **typicality error rate** (TER) is the share choosing the prototypical, wrong image; chance is 0.5. Across languages we also report the cross-lingual flip rate (CFR), stable bias rate (SBR: wrong in every language) and stable correct rate (SCR), which partition items (Appendix B).

Judges. We use Qwen2.5-VL-7B-Instruct (Bai et al., 2025) and InternVL3-8B (Zhu et al., 2025), decoded greedily with a budget of 8 new tokens on images resized to a 512-pixel long side.

4.2 Scorer-based probe: alignment margins

For metric $s(\cdot, \cdot)$ and prompt t_i we take the margin

$$m_i = s(\text{correct}_i, t_i) - s(\text{prototypical}_i, t_i).$$

A negative margin means the metric ranks the prototype higher; the **metric bias rate** is the share with $m_i < 0$ (ties excluded). We use CLIPScore, PickScore and VQAScore, and score the translated prompts with multilingual SigLIP (Zhai et al., 2023).¹

¹Checkpoints: `clip-vit-large-patch14`, `PickScore_v1`, `clip-flant5-xxl` and `google/siglip-base-patch16-256-multilingual`.

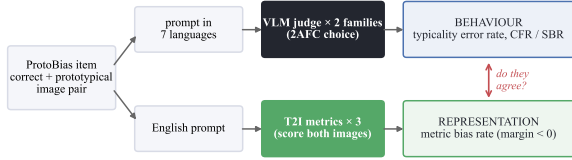


Figure 2: The two measurement paths. The same image pair is judged by two VLM families in seven languages, and scored by three English image–text metrics. The behavioural and representational sides are then compared item by item.

4.3 Statistical analysis

Proportions carry Wilson 95% intervals. On demography we regress the choice on social attribute, prompt language and depicted gender, clustering standard errors by item because each pair appears in seven languages. Likelihood-ratio tests compare each factor, and the attribute $\times$ language interaction, to its reduced model. We report odds ratios; full coefficients and analysis code are in the linked repository.

5 Experiments and results

5.1 Setup and prompt integrity

The run covers $900 \text{ items} \times 7 \text{ languages} \times 2 \text{ families}$: 12,596 judgments on one GPU per model, plus 2,700 English and 6,300 multilingual metric comparisons. Only 4 outputs (0.03%) failed to parse.

Auditing our own pipeline found a defect worth reporting. We had cached translations under the dataset’s `id`, which is not unique: 311 `ids` cover 1,384 rows and 84 span different prompts, putting 256 of the 900 sampled items (28.4%) at risk of being shown a sibling row’s sentence. The dataset also ships a unique `sample_id`; our cache key was the problem. We re-keyed by row index and re-ran everything. In aggregate the defect was almost inert — error rates move by at most 0.009 and every ordering survives — but 11.2% of individual judgments differ between runs, so that agreement is a property of the averages, not the decisions (Appendix D). Running twice also gave us a replicate measurement, used below to separate language from run-to-run noise.

Table 2 reports every language $\times$ domain cell. Both judges are biased overall. Qwen2.5-VL-7B has $\text{TER} = 0.548$ [0.536, 0.561] and InternVL3-8B $\text{TER} = 0.576$ [0.564, 0.588], both excluding chance, but the flat margins hide structure.

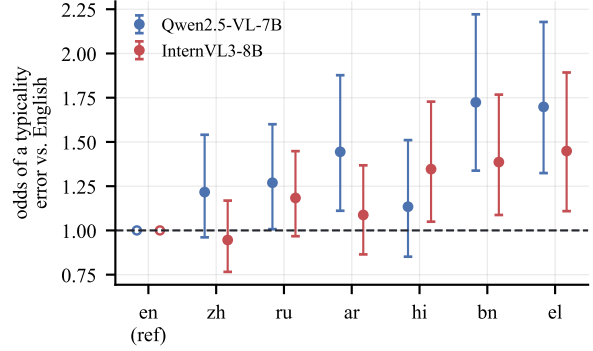


Figure 3: Odds of a typicality error relative to English, from the item-clustered regression; a point above the dashed line means the language elicits more bias. Bengali and Greek sit above it in both families; the intervals are wider for InternVL. Demography items only ($n = 300$), English prompts as the reference level.

5.2 RQ1: changing language changes answers

Asking the same question in a different language moves the answer. Bengali and Greek elicit the most bias in both judges and English the least (Table 2): Qwen ranges from 0.499 in English to 0.598 in Bengali, InternVL from 0.513 in Chinese to 0.648 in Bengali. Relative to English, Bengali raises the odds of a typicality error by 1.72 in Qwen and Greek by 1.70; in InternVL the same three languages lead but more weakly (Greek 1.45, Bengali 1.39, Hindi 1.35; Figure 3).

The regression is more equivocal than the descriptive ordering. On the demography subset the language factor is significant for Qwen ($\chi^2_6 = 17.4$, $p = 0.008$) but only marginal for InternVL ($\chi^2_6 = 11.8$, $p = 0.066$). That subset is 300 items, so the test is not well powered, and we do not claim a significant language effect for InternVL. What both families do show is the descriptive ordering above and the item-level churn below. Our running example is one case: both judges answer correctly in English, Chinese, Russian, Arabic and Greek, and both fail in Hindi and Bengali (Figure 1).

Not explained by measured translation fidelity.

Bengali and Greek have the highest token-F1 in a small back-translation audit; Russian and Hindi score lower (Appendix C). This does not support a simple round-trip-fidelity explanation, but cannot rule out translation effects.

Language-level averages hide item-level churn.

Language-level averages conceal that 68% (Qwen) and 69% (InternVL) of items change their an-

prompt language	Qwen2.5-VL-7B			InternVL3-8B		
	animal	object	demog.	animal	object	demog.
English	0.533	0.447	0.517	0.597	0.490	0.543
Chinese	0.570	0.452	0.563	0.560	0.450	0.530
Russian	0.550	0.492	0.573	0.603	0.503	0.583
Arabic	0.580	0.450	0.603	0.570	0.480	0.563
Hindi	0.597	0.517	0.547	0.587	0.593	0.613
Bengali	0.603	0.547	0.643	0.697	0.627	0.620
Greek	0.577	0.517	0.640	0.670	0.590	0.630
all languages	0.573	0.489	0.584	0.612	0.533	0.583

Table 2: Typicality error rate, the share of choices landing on the prototypical but incorrect image, by language and domain over all 900 items. 0.5 is chance.

swer in at least one language. What *is* stable is the strongly biased material — wealth items are answered prototypically in every language about three times as often as intellect items are (Table 3).

How much of that churn is noise? A judgment that changes between languages is only interesting if the same judgment would not change on its own. The re-run gives us that baseline for free: on the items whose prompt was never affected by the cache defect, the two runs are a replicate measurement under an identical prompt and an identical image pair. InternVL reproduces itself almost exactly — 1.1% of its judgments differ between runs — so its cross-language disagreement of 0.298 is 26 times its own noise floor. Qwen is far less stable: 17.1% of its judgments flip on re-run under greedy decoding, against a cross-language disagreement of 0.288, a ratio of only 1.7. Language moves both models more than chance does, but for Qwen a substantial share of the 68% churn is decoding instability rather than language, and its item-level numbers should be read with that in mind. We report the ratio because the raw churn figure alone would overstate the effect.

5.3 RQ2: no attribute-profile change is detected

The overall averages hide a large semantic contrast. Both judges make many more errors on wealth than on the least biased attribute, a spread of 0.29 (Qwen, morality 0.496 to wealth 0.783) and 0.31 (InternVL, intellect 0.482 to wealth 0.793; Figure 4a). Social attribute is significant in both families ($\chi^2_4 = 81.8$ and 71.7 , both $p < 10^{-14}$); wealth carries 3.8 (Qwen) and 3.5 (InternVL) times the error odds of morality.

Changing language shifts these levels but does not measurably redraw the profile: the attribute

$\times$ language interaction reaches significance in neither family ($p = 0.055$ for Qwen, $p = 0.87$ for InternVL). Qwen’s value is close enough to threshold that we read it as untested rather than absent. The other domains also show contrasts: birds and generic animals exceed mammals, and vehicles exceed furniture. This is consistent with, but does not establish, a role for canonical visual form.

5.4 RQ3: the model families partly agree

The two models’ per-item error rates correlate at $r = 0.69$; their individual decisions have Cohen’s $\kappa = 0.43$. This is consistent with a substantial item component, but leaves substantial model-specific or unexplained variation.

5.5 RQ4: the category pattern also appears in scorer outputs

Without the two-image forced-choice judge, CLIP-Score ranks the prototype above the correct image on 0.710 of items and PickScore on 0.673; VQAScore has a lower rate of 0.472.

Every scorer ranks the prototype above the correct image more often on wealth than intellect, including near-chance VQAScore (Figure 5a). Thus the contrast appears in independent scorer outputs as well as both VLM judges.

VLM choices also track scores item by item (Figure 5b): the correct-pick gap is 0.25–0.35 across all six scorer–model combinations. This association does not establish a shared causal mechanism.

The behavioural language ordering is not reproduced by SigLIP. Multilingual SigLIP ranges from 0.579 to 0.661, and its lowest language is Bengali (Figure 5c) — the language that elicits the *most* bias from both judges. It does not

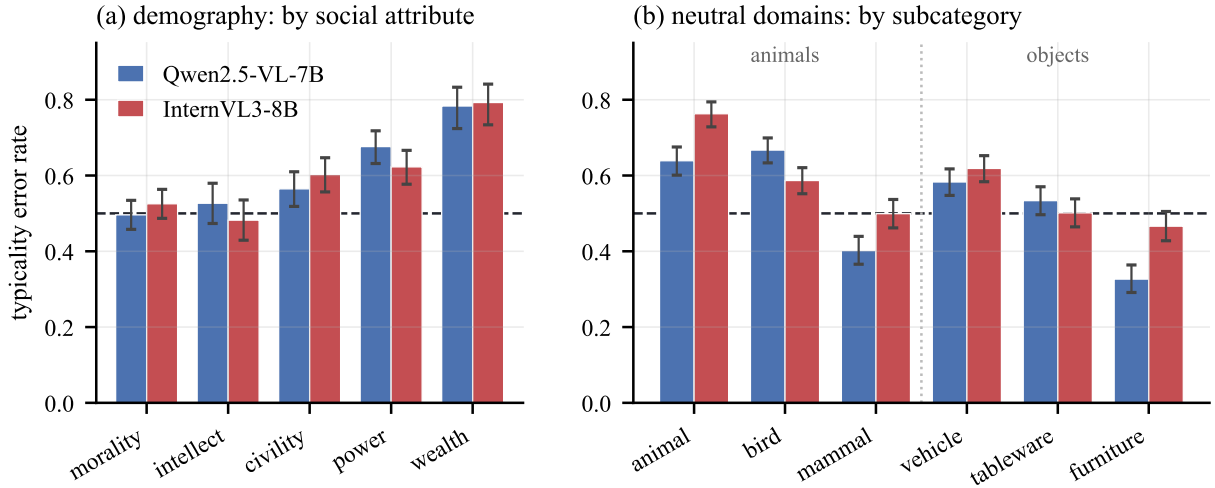


Figure 4: Error rates vary by semantic category, pooled over all seven prompt languages and both judges. (a) Wealth and power show more errors than intellect. (b) Related contrasts appear outside demography. Wilson 95% intervals; dashed line is chance.

reproduce the behavioural ordering. One encoder cannot locate where that effect arises.

5.6 Ablations and confound checks

Each prototypical image perturbs exactly one visual property, so we can ask which perturbations the judges miss (Figure 6). If the effect were a colour artifact, colour should be the worst cue; instead it is the *best* (0.404 and 0.482, the only cue at or below chance). Scale has the highest error rate in both models (0.664/0.687), followed by count (0.626 in both).

The images are generated, and the atypical instance is the harder one to draw: an octopus sits further from a text-to-image model’s comfort zone than a dog. If the judges were simply avoiding the worse-rendered picture, that alone could produce our results. The benchmark ships a per-image quality score from two large VLM filters, and the asymmetry is real — the correct image averages 9.75 against 9.96, and in 32% of items the prototypical image scores higher against 2% the other way. It does not track the errors. The correlation between the quality gap and the item error rate is +0.05, and the pattern across cues runs backwards: colour has one of the largest gaps and the *lowest* error rate, scale the smallest gap and the *highest*. The benchmark’s weakest generator also yields slightly less bias than its strongest. These scores come from 32–38B models of the families we evaluate and sit near ceiling, so they bound the explanation rather than eliminate it.

Finally, Qwen answers “1” on 0.473 of trials

and InternVL on 0.543. Because the correct image is randomised across slots, neither frequency can produce the aggregate, and wealth exceeds morality within either slot.

5.7 Error analysis

Items sort into a stable core and a language-sensitive middle. Both judges are correct in all 14 conditions on 31 items and wrong in all 14 on 114; the remaining 755 change somewhere. A platypus–dog item turning on a branch in the foreground versus the background is answered correctly everywhere; a walking stick insect–monkey item turning on a red leaf pile is answered prototypically everywhere.

The composition of those two groups is informative. Colour supplies 15 of the 31 always-correct items but only 14 of the 114 always-wrong ones, which are led instead by spatial (31), count (26) and scale (24). Colour is both the cue a text-to-image model renders most reliably and the cue our judges handle best, so the always-wrong group plausibly mixes genuine bias with items whose manipulation is not clearly visible (§6). We cannot separate the two without human annotation. Finally, wealth is the most stable failure but also the smallest cell ($n = 31$ items), so its magnitude is the least certain number we report.

6 Conclusion

Holding the image pair fixed and changing only the language of the question, we find that a vision–language judge is not one scoring function but sev-

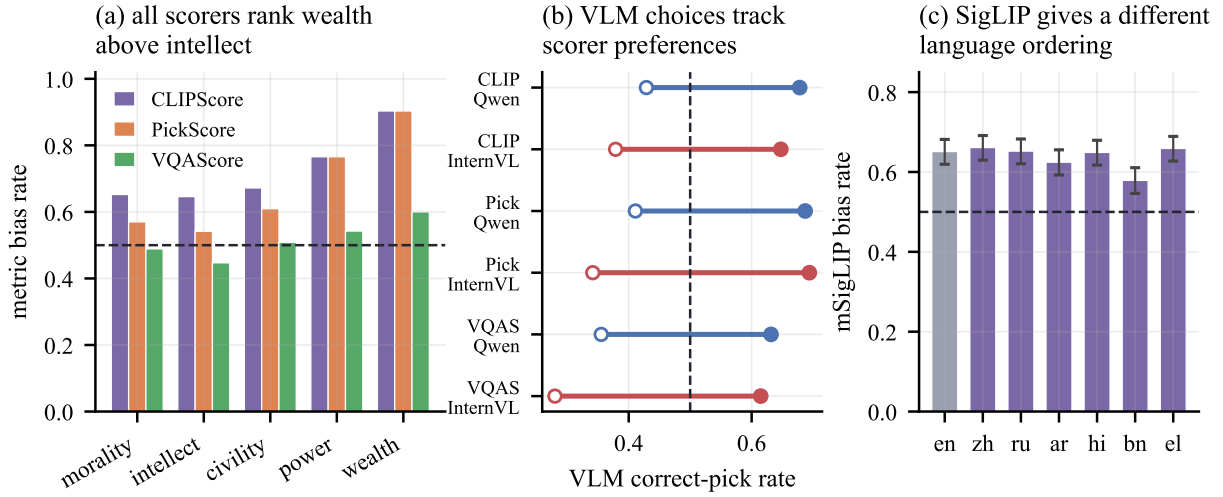


Figure 5: Scorer behaviour. **Panels (a) and (b) use English prompts only**, because CLIPScore, PickScore and VQAScore are scored on the English caption. (a) All three scorers place wealth above intellect, including VQAScore, whose overall bias rate is near chance. (b) Filled marker: VLM correct-pick rate when the scorer prefers the correct image; open marker: when it prefers the prototypical one. **Panel (c) uses all seven languages**: multilingual SigLIP scorer bias ranges from 0.579 to 0.661, with Bengali lowest.

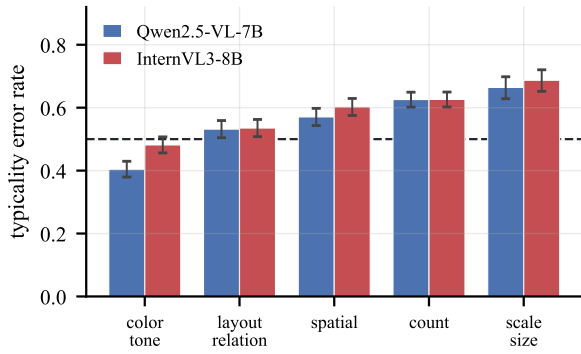


Figure 6: Error rate by the single visual property manipulated to build the prototypical image, pooled over all seven prompt languages. Colour, the obvious artifact hypothesis, is the *least* biased cue.

eral. The error rate rises with the prompt language — English lowest, Bengali and Greek highest in both families — and roughly two thirds of individual items change verdict in at least one of seven languages. The regression confirms this for Qwen and leaves it marginal for InternVL, so we report the effect as clear descriptively and unevenly supported statistically. What does not move is the *shape* of the bias: both families are most vulnerable on wealth by a wide margin and near chance on their least biased attribute, with no attribute-by-language interaction. That profile is not confined to the judges. Three independent image–text scorers reproduce the wealth contrast without being asked a question, which places the category structure in the shared representation rather than

in any one model’s decoding. The language effect does not follow: a multilingual encoder shows bias in every language but ranks them differently, with Bengali its *least* biased language and the judges’ most.

The design is what makes the language claim robust to the benchmark’s weakest point. Its images are generated, and some pairs do not visibly realise the manipulation they describe, which inflates absolute error rates by an unknown amount — but every language is asked about the identical pair, so image quality cannot create a difference between languages. Prototypicality bias is therefore a cross-lingual evaluation problem, not only an image-scoring one: a judge validated in English carries neither its error rate nor its item-level decisions to another language, so multilingual pipelines should report per language and per category rather than in aggregate.

Limitations

The most serious limitation is item validity. The benchmark’s images are generated, and inspecting pairs by hand shows that the manipulated cue is not always visible in them: a caption may contrast a mound behind the animal with one in front of it while the two images differ in no way a reader can name, and some objects are rendered incoherently enough that it is unclear what they are. Our data carry an indirect trace of this. Colour, the cue a text-to-image model places most reliably, sup-

plies half the items both judges always get right but only an eighth of the items they always get wrong, which are led by the spatial, count and scale manipulations that are hardest to render unambiguously. Some part of the error rates we report is therefore item noise rather than model bias, and this bounds the absolute numbers throughout. It does not touch the comparisons: every language sees the identical image pair, so no property of the images — rendering quality, an unrealised manipulation, or anything else — can produce a difference between languages. Comparisons across cue types are more exposed, and we treat them as descriptive.

Establishing how much is item noise needs human annotation, which we designed but did not run. The interface, the sampling script and the counterbalanced multilingual protocol are released as `exp3j_human_eval`: each annotator judges half the items in their own language and half in English, so that a per-annotator English baseline and an inter-annotator agreement estimate come out of the same session. A study using it would put a floor under the judges’ error rates.

Second, the statistical evidence for the language effect is uneven. It is clear in the descriptive rates and in the item-level churn, but the regression is fit on the 300 demography items and reaches significance for Qwen only. Third, wealth rests on 31 items with a wide odds-ratio interval and one sampling seed. Run-to-run variation we can now quantify, and for Qwen it is large: 17.1% of its judgments flip when the identical prompt and image pair are evaluated a second time, so its item-level results carry much more uncertainty than the interval on any single rate suggests. InternVL flips on 1.1%. Fourth, the seven languages confound resource level, script and culture, so they demonstrate the effect without explaining it. Finally, typicality and the manipulated cue are not independently crossed, only 7–8B judges are tested, and the cross-lingual scorer result rests on a single multilingual encoder.

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A Contribution statement

This is a single-author submission, and I am the sole human author responsible for this report. I designed the cross-lingual extension, selected the languages, and held the images fixed while changing the linguistic input. With the AI assistance in Appendix F, I developed the 2AFC pipeline, translation and model backends, slot randomisation and cluster jobs; ran Qwen and InternVL; and implemented the clustered regression, stability, visual-cue, cross-model and scorer analyses. I also performed the prompt-integrity audit, designed the pending human study and interface, generated the figures and tables, and wrote and revised the report. Earlier progress reports were presented jointly with another student who is not an author of this paper. This report follows the single-author arrangement, and I take responsibility for its claims.

B Cross-lingual stability

Table 3 summarises each item *across* its languages rather than per judgment: whether its verdict is constant, and if so in which direction. The three rates partition the items and sum to one.

model	subset	CFR	SBR	SCR
Qwen2.5-VL-7B	all items	0.677	0.192	0.131
	wealth	0.323	0.581	0.097
	power	0.703	0.219	0.078
	morality	0.677	0.151	0.172
	intellect	0.792	0.125	0.083
InternVL3-8B	all items	0.690	0.230	0.080
	wealth	0.516	0.452	0.032
	power	0.516	0.344	0.141
	morality	0.645	0.204	0.151
	intellect	0.688	0.188	0.125

Table 3: Per-item stability across languages. CFR: the answer changes in at least one language. SBR: prototypical in *every* language. SCR: correct in every language. The strongly biased attributes are also the most stably wrong.

C Translation quality audit

For 40 prompts per language, same-system back-translation token-F1 ranges from 0.785 (Russian) to 0.880 (Greek); Bengali scores 0.858, and no prompt falls below 0.5. This small round trip measures consistency, not human translation quality. Exact prompts and templates are in the repository.

D Effect of the prompt-integrity fix

We ran the full evaluation twice: once with the defective `id`-keyed translation cache and once after re-keying by row index. Aggregate results are nearly unchanged. Overall error moves from 0.548 to 0.548 for Qwen and from 0.585 to 0.576 for InternVL; English and Chinese remain the least biased languages and Bengali and Greek the most; the wealth peak holds at 0.783 and 0.793.

Individual judgments are another matter: 11.2% of them differ between the two runs. Splitting by whether the item’s prompt could have been affected shows both causes at work. For items the defect could not touch, 17.1% of Qwen’s judgments and 1.1% of InternVL’s still flip, which is decoding instability rather than the fix. For items at risk, the rates rise to 23.7% and 9.7%. The excess — roughly 7 and 9 percentage points — is the fix doing its work. This is why the aggregate agreement between the two runs should not be read as evidence that the defect was harmless.

E Prompt templates

Figure 7 gives the two-image choice instruction in all seven languages, exactly as sent to the judges. `{desc}` is replaced by the translated neutral description of the item. Every template ends by requiring a single ASCII digit, 1 or 2, so that answer parsing is identical across scripts; the digit refers to the image slot, and the correct image is randomised between slots per item.

The English, Chinese, Arabic and Hindi templates were written by hand. The Russian, Bengali and Greek templates were drafted with LLM assistance, which is worth checking, because Bengali and Greek are the two languages with the highest error rates: an awkward instruction, rather than the language itself, could explain their position. Round-tripping each template through back-translation does not support that. All six non-English templates return the English instruction with no change of meaning — Bengali gives “See Figure 1 and Figure 2. Which image best matches this description: ... Answer with only one number: 1 or 2” and Greek “Look at Picture 1 and Picture 2 ... Answer with only one digit: 1 or 2” — and the two LLM-drafted high-error templates round-trip no worse than the hand-written ones. The instruction wrapper is therefore not a plausible source of the language ordering. Note

ENGLISH (EN)
Look at Image 1 and Image 2. Which image better matches this description: "{desc}"? Answer with only the single digit 1 or 2.
CHINESE (ZH)
请看图1和图2。哪张图更符合下面这句描述：“{desc}”？只回答数字 1 或 2。
RUSSIAN (RU)
Посмотрите на Изображение 1 и Изображение 2. Какое изображение лучше соответствует этому описанию: "{desc}"? Ответьте только одной цифрой: 1 или 2.
ARABIC (AR)
انظر إلى الصورة 1 والصورة 2. أي صورة تطابق هذا الوصف بشكل أفضل: "{desc}"؟ أجب برقم واحد فقط 1 أو 2.
HINDI (HI)
चित्र 1 और चित्र 2 देखें। कौन सा चित्र इस विवरण से बेहतर मेल खाता है: "{desc}"? केवल एक अंक 1 या 2 में उत्तर दें।
BENGALI (BN)
চিত্র 1 এবং চিত্র 2 দেখুন। কোন চিত্রটি এই বিবরণের সাথে বেশি মিলে: "{desc}"? শুধুমাত্র একটি সংখ্যা দিয়ে উত্তর দিন: 1 বা 2।
GREEK (EL)
Κοιτάξτε την Εικόνα 1 και την Εικόνα 2. Ποια εικόνα ταιριάζει καλύτερα με αυτήν την περιγραφή: "{desc}"; Απαντήστε μόνο με ένα ψηφίο: 1 ή 2.

Figure 7: The two-image choice instruction in all seven languages, with the required output format. Rendered as an image so that every script displays correctly.

that this checks the wrapper, not the item descriptions, which are audited separately in Appendix C.

F AI usage declaration

I used AI assistants, including Claude and Codex, to help develop and debug the evaluation pipeline, analysis scripts, experiment tooling, figures and tables, and to edit the report. Examples include the SLURM scripts, plotting functions, the $\text{\LaTeX}$ table generator and the item-clustered NUMPY/SCIPY analysis. For the report, Codex shortened an earlier draft, moved detailed material to the appendix, audited claims against the saved outputs and checked the rendered pages.

I did not treat model-generated prose or unverified AI answers as experimental evidence. The reported numbers come from the saved predictions and metric scores. Figures and tables are generated by `make_figures.py` and `make_tables.py`. I checked quantitative claims against these files after editing the text.

F.1 Concrete examples

VQAScore implementation

The `t2v_metrics` package that implements VQAScore pins its own dependencies, so I first developed a self-contained backend with AI assistance using BLIP-2, whose language model is FlanT5. Several plausible scoring implementations produced near-uniform or image-insensitive outputs. I tested decoder-forward, generated-token and forced-answer readouts, and also changed the model dtype from `float16` to `bfloat16`. Because these diagnostics did not establish a reliable continuous score for the fallback, I did not use it in the reported analysis. The VQAScore results in this paper come from the package default `clip-flanT5-xxl`, run separately on the A40 cluster. This episode taught me that a metric completing without an exception is not enough; its output must also pass sanity checks.

The dataset key

An early AI-assisted version of the translation cache used `translations[row["id"]][lang]`. I did not initially verify that `id` was unique. During the paper audit I checked this assumption and found that 311 ids cover 1,384 rows. Some rows with the same id even have different prompts. I re-keyed the cache by row index and re-ran the full evaluation, and I report the defect in the paper rather than only the corrected numbers, because a reader cannot otherwise tell a checked

pipeline from an unchecked one.

Choosing the aggregation

An early analysis aggregated errors by domain and language. This gave a fairly flat demography result of about 0.56. I later split the same rows by `socio_attr`, which exposed the large difference between intellect and wealth that became the main result. The original aggregation was not technically wrong, but it was too coarse for the research question.

F.2 Responsibility

AI-generated code was useful as a starting point, but it sometimes made reasonable-looking assumptions that were wrong for this dataset. I therefore reran the analyses from the saved data, compared unexpected results with separate checks and audited the final wording against the available evidence. The report combines my project notes and earlier drafts with AI-assisted editing. I take responsibility for the experimental design, interpretation and any remaining errors.